

① Goal: SHEN - SHORNAKER

2 SPEC & Γ -STRUCTURE FOR STANDARD FLIPS

Γ -Conj 2: X Fano

$$D^b(X) = \langle E_1, \dots, E_n \rangle \text{ FEC}$$

$$K_0(X) = \langle [E_1], \dots, [E_n] \rangle \text{ SOB}$$

$$\hat{\Gamma}_X \circ \text{Ch}(-) : K_0(X) \xrightarrow{\sim} H^*(X, \mathbb{C})$$

$$H^*(X, \mathbb{C}) \cong \hat{\Gamma}_X \circ \text{Ch}(E_i) - \text{SOB}$$

Asymptotic classes:

Flat sections on $H^*(X, \mathbb{C}) \otimes \partial_{p_1}$ on $\mathbb{P}^1 \setminus \{0, \infty\}$

0-irreg singular

$\lambda_1, \dots, \lambda_n$ - eigenvalues of $c_1(X)_0^* - \text{on } H^*(X, \mathbb{C})$

$\exists$ flat sections s_i ($i=1, \dots, n$) on sectors

including $\lambda_i \mathbb{R}_{>0}$ st $\|s_i\| \sim e^{-\lambda_i t} z$ at $z \rightarrow 0$

$s_i \xrightarrow{z \rightarrow \infty} A_i \in H^*(X, \mathbb{C})$ at $z \rightarrow \infty$ $z \in \lambda_i \mathbb{R}_{>0}$

Conj: $A_i = \hat{\Gamma}_X \circ \text{Ch}(E_i)$ for some FEC in $D^b(X)$

Goal: For certain SOD from bir geom

$$D^b(X) = \langle \mathcal{A}_1, \dots, \mathcal{A}_n \rangle \quad \mathcal{A}_i \cong D^b(Y_i)$$

$$K_0(X) = K_0(\mathcal{A}_1) \oplus \dots \oplus K_0(\mathcal{A}_n)$$

$$\hat{\Gamma}_X \circ \text{Ch}(-) \quad H^*(X, \mathbb{C}) = \bigoplus H^*(Y_i, \mathbb{C})$$

Explanation of this via asymptotic
claims for "extremal" QH on $H^*(X, \mathbb{C})$

Today: SOD for $\mathbb{P}^n$

Proj. bundles

Blow-ups

Standard flips

(Birational transformations)

SOD: $D^b(X) = \langle \mathcal{A}_1, \dots, \mathcal{A}_n \rangle$ $\mathcal{A}_i$ full s-d
 $\text{Hom}^*(\mathcal{A}_j, \mathcal{A}_i) = 0 \text{ } \forall j > i$

and $\mathcal{A}_1, \dots, \mathcal{A}_n$ generate $D^b(X)$
 $(\Leftarrow) \forall F \exists \bigoplus_{i=1}^n \mathcal{A}_i \rightarrow F \rightarrow F_{n+1} \rightarrow \dots \rightarrow F_1 = F$
 $\exists! \text{ Cone}(F_{k+1} \rightarrow F_k) \in \mathcal{A}_k$

FEC $E_1 \dots E_n \leadsto \mathcal{A}_i = \langle E_i \rangle \cong D^b(\text{1-vec})$
 $\cong D^b(\text{pt})$

For $\mathbb{P}^n$ $(\mathcal{O}(a), \mathcal{O}(a+1), \dots, \mathcal{O}(a+n))$ - FEC
 $\forall a \in \mathbb{Z}$

Th. Orlov Y , E-v.b. on Y $rk=r$
 $X = \mathbb{P}(E) \xrightarrow{p} Y \quad \mathcal{O}_X(1)$

$\forall a \quad D^b(X) = \langle D_a, D_{a+1}, \dots, D_{a+r-1} \rangle$

where $D_i = Lp^*(\) \otimes_{\mathcal{O}_X} \mathcal{O}(i) \cong D^b(Y)$

Lemma $p: X \rightarrow Y$ see proj $Rp_* \mathcal{O}_X \cong \mathcal{O}_Y$
 Then $Lp^*: D^b(Y) \rightarrow D^b(X)$ fully faithful
Pr
 $\text{Hom}_{D^b(X)}(Lp^* F_1, Lp^* F_2) \cong \text{Hom}_{D^b(Y)}(F_1, Rp_* Lp^* F_2) \cong$
 $\cong \text{Hom}(F_1, F_2 \otimes^L Rp_* \mathcal{O}_X) \cong$
 $\cong \text{Hom}(F_1, F_2)$

Pr
 $\text{Hom}(Lp^* F_1 \otimes \mathcal{O}(j), Lp^* F_2 \otimes \mathcal{O}(i)) =$
 $= \text{Hom}(Lp^* F_1, Lp^* F_2 \otimes \mathcal{O}(i-j))$
 $= \text{Hom}(F_1, (Rp_* Lp^* F_2 \otimes \mathcal{O}(i-j))) \cong$
 $\cong \text{Hom}(F_1, F_2 \otimes^L Rp_* \mathcal{O}_X(i-j))$
 $Rp_* Lp^* F_2 \otimes^L \mathcal{O}(i-j) \cong F_2 \otimes^L Rp_* \mathcal{O}_X(i-j) = 0 \quad -r < i-j < 0$

$$D^b(x) = \langle \langle x_1, \dots, x_n \rangle^\perp, x_1, \dots, x_n \rangle \text{ -- SOD}$$

$$= \langle x_1, \dots, x_n, \langle x_1, \dots, x_n \rangle^\perp \rangle$$

$$\text{Hom}(F, R_{P_2}(G \otimes O(-i))) = 0$$

↓

~~SECRET~~

$$\begin{array}{ccc} Y \times \mathbb{P}^{n-1} & \xrightarrow{q} & \mathbb{P}^{n-1} \\ \downarrow p & & \downarrow p \\ Y & \xrightarrow{\quad} & \mathbb{P}^1 \end{array}$$

$$R_{\phi} G = 0$$

$$G = 0$$

$$P_N = \tilde{Z} \xrightarrow{i} \tilde{X} = B|_{\tilde{Z}} X$$

$$\frac{Z \rightarrow X}{\quad}$$

$$\textcircled{b}) L_P^* : D^p(x) \rightarrow D^p(\bar{x}) \text{ f.f.}$$

$$\tilde{D}_i = i_\infty D_i$$

$$D_i = (\phi^{-1} D^b \hat{Z}) \otimes \mathcal{O}(i)$$

$$\text{Hom}(F_1 \otimes_{\mathbb{Z}} (1), F_2) = 0$$

for z $F_{j,z} \in D_i$
 $F_z \in D_j$

if ~~$j \in [0, \dots, r-2]$~~
 $j - (i+1) \in [r+1, \dots, -1]$

$$j-i \in \{1, 2, \dots, n\}$$

~~Standard flip~~ ~~Standard flip~~ Z $V_1, V_2 - v.b.$
 $rk = r, s$

$$F = P(V_1)$$

$$F' = P(V_2)$$

$$E = P \times_F F'$$

$$X \supset F \quad N_{F/Z} \cong \mathcal{O}(-1) \otimes \pi^* V_2 \cong \mathcal{O}(-1)^s$$

$$E \hookrightarrow B_F X = \widehat{X} = B_{F'} X' \xrightarrow{i} E$$

$$\swarrow \pi \quad \searrow \pi \quad \searrow P(\mathcal{O}(-1) \otimes \pi^* V_1)$$

$$F \xrightarrow{i} X \xrightarrow{\phi} X' \xleftarrow{i} F'$$

$$\searrow P(V_1) \quad \swarrow P(V_2)$$

$$Z$$

$r \geq s$ flip
 $r = s$ flop

Th B-O, Beltramini-Raedschelders:

$$D^b(X) = \langle D_{s-r}, \dots, D_{-1}, \pi_* \pi^* D^b(X') \rangle$$

where $D_i = j_* (\pi^* D^b(Z) \otimes \mathcal{O}_F(i))$

Ex $A^r \times A^s$ G_m on Y

$$t(x_1, \dots, x_r, y_1, \dots, y_s) = (tx_1, \dots, tx_r, t^r y_1, \dots, t^r y_s)$$

$$R = k[x, y]^{G_m} = k[x, y^r]$$

$$\text{Spec } R = (A^r \times A^s) //_{G_m} = \text{Grass} (P^{n-1} \times P^{s-1} \rightarrow P^{r+s-1})$$

$$X = A //_{G_m} = (A^r \setminus 0) \times A^s //_{G_m} \rightarrow P^{n-1}$$

$$\parallel$$

$$\text{Tot}(\mathcal{O}(-1)^s) \xleftarrow{P^{n-1}} \xrightarrow{F}$$

$$X' = A //_{G_m} = (A^r) \times (A^s \setminus 0) //_{G_m} \rightarrow P^{s-1}$$

$$\text{Tot} \mathcal{O}(-1)^r \xleftarrow{P^{s-1}} \xrightarrow{F'}$$

$$\widehat{X} \supset P^{n-1} \times P^{s-1}$$

$$\swarrow \quad \searrow$$

$$P^{n-1} \supset X \quad X' \supset P^{s-1}$$

$$\searrow \quad \swarrow$$

$$S \supset B$$